

Pradeep Kumar Vijayakumaran

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Summary

AI & Data Engineer with **2+** years experience building scalable data pipelines, LLM retrieval systems, and cloud-native platforms for clients like **Nike** and **L'Oréal**. Proficient in production ETL/ELT, vector search, and model observability on AWS and GCP, focused on reliability and cost efficiency. Pursuing an MSCS at **Northeastern University**, specializing in ML, agentic AI, and distributed data systems.

Experience

Fashion Index

Digital Cataloguing Intern

Charleston, SC

Jan. 2026 – Apr. 2026

- **Achieved 95%+ visual search accuracy** for a pre-launch platform by building a hybrid vector search system combining 384-dimensional DINOv2 embeddings with custom L2 distance scoring for dominant-color extraction
- **Scaled platform inventory to 56k+ products** by engineering an operational, stateless ETL ingestion pipeline across 15+ suppliers, leveraging a MongoDB task queue and direct-to-S3 streaming prior to public launch
- **Engineered a pre-launch e-commerce storefront and admin panel** in React.js backed by FastAPI, enabling voice-driven catalog search and real-time execution log streaming

Infosys Limited

Senior Systems Engineer, Data and Analytics

Karnataka, India

Oct. 2021 – Jun. 2022

- **Maintained 99.9% uptime across L'Oréal's** mission-critical analytics systems by engineering proactive monitoring and automated alerting workflows, eliminating recurring SLA violations
- **Accelerated data ingestion throughput by 3x** across major Japanese retailer data feeds by designing production Scala and T-SQL (SSMS) pipelines with Control-M scheduling
- **Reduced MTTR by 50%** during a high-risk on-premises to GCP migration by building real-time alerting APIs and automated Git-based CI/CD incident response

Systems Engineer, Data and Analytics

Oct. 2020 – Sep. 2021

- **Delivered 100% data fidelity with zero downtime** for Nike Inc. by executing enterprise platform modernization and migration from SAP AFS to SAP S/4HANA
- **Eliminated downstream data gaps** for analytics teams by rebuilding legacy ingestion into fault-tolerant distributed ETL pipelines using Python, Spark, Airflow, and AWS
- **Lowered AWS infrastructure costs by 40%** by profiling Airflow DAG resource usage, rightsizing over-provisioned EC2/EMR clusters, and enforcing automated infrastructure alerting

Data Analyst Intern

Jan. 2020 – Mar. 2020

- **Earned 'Top Performer' distinction** by delivering end-to-end ML and analytics applications using Python (NumPy, Pandas, and scikit-learn) and SQL for data modeling, predictive analysis, and visualization

Projects

VERITAS: Retrieval Grounding & Verification Engine | *Python, FastAPI, Streamlit, Anthropic API, Pinecone, AWS S3, Airflow, Docker*

- Eliminated **98%** of hallucinations across 200+ claims by building a multi-agent RAG verifier auditing arXiv and web sources
- Accelerated doc indexing throughput by **4x** by orchestrating layout-aware Airflow and AWS S3 ETL pipelines into Pinecone

VERTO: Natural Language Database Interface Engine | *Python, PyTorch, PEFT/LoRA, T5, FastAPI, SQLite, Spider*

- Achieved **99%** valid SQL query generation on 1,034 Spider dev queries by fine-tuning T5-large with LoRA adapters
- Reduced prompt token usage by **45%** via context-aware schema-linking heuristics and closed-loop SQL self-healing recovery

VANTAGE: LLM Telemetry & Observability Platform | *Python, Streamlit, SQLite, SQLAlchemy, Anthropic API, Plotly*

- Engineered **sub-2ms** tracing for nested agent trees using thread-safe context variables across 10,000+ execution spans
- Automated prompt drift detection across **500+** daily runs using LLM-as-a-judge scoring and Z-score anomaly alerts in Plotly

Skills

Languages & Analytics: Python, SQL, JavaScript, Scala, R, DuckDB, Plotly, Power BI, Tableau

Data Engineering & AI: PyTorch, Hugging Face (PEFT/LoRA), Pinecone, Airflow, dbt, FastAPI, Spark, Polars, scikit-learn, LangChain

Cloud & Infrastructure: AWS (S3, ECS), GCP, Snowflake, SQLite, MySQL, Supabase, Docker, Kubernetes, Terraform, Git, CI/CD

Education

Northeastern University

Master of Science in Computer Science

Boston, MA

Sep. 2024 – Dec. 2026

Anna University

Bachelor of Engineering in Computer Science and Engineering

Tamil Nadu, India

Aug. 2016 – Sep. 2020